

Foundation (Jalapeno)

- Q1.** Solve the inequality $|x| < 3$.
 (A) $(-3, 3)$
 (B) $(-\infty, -3) \cup (3, \infty)$
 (C) $[-3, 3]$
 (D) $(-\infty, -3] \cup [3, \infty)$
 (E) $(-3, 3]$
- Q2.** Solve the inequality $|2x - 1| \geq 5$.
 (A) $(-\infty, -2] \cup [3, \infty)$
 (B) $(-\infty, -2) \cup (3, \infty)$
 (C) $(-2, 3)$
 (D) $[-2, 3]$
 (E) $(-\infty, 3] \cup [-2, \infty)$
- Q3.** Graph the solution set of $|x + 2| < 4$ on a number line.
 (A) $(-6, 2)$
 (B) $(-6, 2]$
 (C) $[-6, 2)$
 (D) $[-6, 2]$
 (E) $(-2, 6)$
- Q4.** The temperature in a city is never more than 25°C above or below 0°C . Find the range of possible temperatures using absolute value inequality.
 (A) $|-25| \leq 0$
 (B) $|0 - 25| \leq t$
 (C) $|t| \leq 25$
 (D) $|t - 0| \geq 25$
 (E) $t \leq 25$
- Q5.** Solve the inequality $|x - 4| > 2$.
 (A) $(-\infty, 2) \cup (6, \infty)$
 (B) $(-\infty, 2] \cup [6, \infty)$
 (C) $(-\infty, 2) \cup [6, \infty)$
- (D) $(-2, 6)$
 (E) $[-2, 6]$
- Q6.** Express the solution set of $|3x - 2| \leq 1$ in interval notation.
 (A) $\left[-\frac{1}{3}, \frac{1}{3}\right]$
 (B) $\left[-\frac{1}{3}, \frac{1}{3}\right)$
 (C) $\left(-\frac{1}{3}, \frac{1}{3}\right]$
 (D) $\left(-\frac{1}{3}, \frac{1}{3}\right)$
 (E) $\left[-\frac{1}{3}, \frac{1}{3}\right]$
- Q7.** Find the values of x for which $|x + 5| < |x - 3|$.
 (A) $(-\infty, -1)$
 (B) $(-1, \infty)$
 (C) $(-5, 3)$
 (D) $(-5, -1)$
 (E) $(-1, 3)$
- Q8.** A company packages nuts in bags that weigh within 0.1 pound of the average weight. If x represents the weight of a bag of nuts, the absolute value inequality that models this situation is
 (A) $|x| \leq 0.1$
 (B) $|x - 0.1| \leq 0.1$
 (C) $|x - 0| \leq 0.1$
 (D) $|x - 0.1| < 0.1$
 (E) $|x| \geq 0.1$

Open Ended Questions (FRQ)

- Q9.** A baker sells a specialty bread that weighs within 0.05 pound of the average weight of 1.5 pounds. Write an absolute value inequality that represents the range of possible weights of the bread. Solve the inequality and express the solution in interval notation.
- Q10.** The distance between the points $(-2, 3)$ and $(4, -1)$ on the coordinate plane is given by $|x_2 - x_1| + |y_2 - y_1| = 9$. Find all possible values of x and y that satisfy this equation, where the distance between two points (x_1, y_1) and (x_2, y_2) is given by the equation. Show the solution set on a coordinate plane and express it in interval notation.

Chef's Tips

- Tip 1: Remind students to check their solutions by plugging the values back into the original inequalities.
- Tip 2: Provide extra support to students who struggle with graphing the solution sets on the number line.
- Tip 3: Consider using real-world applications and word problems to help students better understand the concept of absolute value inequalities.